Battery and Power Management

(Embedded System Perspective)

Introduction

- Batteries are cells which can generate electricity.

- A battery can have one or more electrochemical pack of cells arranged in a series/parallel fashion so that the voltage can be raised to desired levels.

- Primary cells and secondary cells are two types of batteries.

- The major difference between a primary battery and the secondary battery is that primary cells are the ones that are non-rechargeable but secondary cells are the ones that are rechargeable.

- Why DC is preferred over AC?

Terminologies used while dealing with batteries

Unique terms that defines the characteristics of a battery:

- ❑ Power capacity.
- ❑ C-rating.
- ❑ Nominal voltage.
- ❑ Charging voltage.
- ❑ Charging current.
- ❑ Discharging current.
- ❑ Cut off voltage.
- ❑ Shelf life, cycle .

Power capacity

- It is the **energy stored in a battery** which is measured in **Watt-hour.**

- We generally see the battery ratings as 2500mAh or 4000mAh while reading the specifications of a smart phone.

Power capacity

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- Watt-hour = V * I * hours {since voltage is kept constant, so it is measured in Ah/mAh}
- We generally see the battery ratings as 2500mAh or 4000mAh while reading the specifications of a smart phone.

- **Example:** *2500 mAh it means that the battery has a capability to deliver 2.5A/2500mA of current to the load for 1 hour. The time that the battery works continuously depends upon the load current that it consumes.*
- *So if the load consumes only 25 mA of current then the battery can stay alive for 100 hours. How is it?*

Practically battery's duration changes based on the **temperature and the **current** consumption etc.*

Power capability(C-rating)

- It means the **amount of current that the battery can deliver**. It is also known as **C-rating.**
- Theoretically, it is calculated as A-h divided by 1 hour.
- If a battery has 1C rating then the battery has a capability of delivering 1A of current.
- ***Note:*** Higher the C rating, more the current that can be drawn from the battery.
  - **Example**: Let's consider a battery which has 10000 mAh of power capacity.

Power capability(Contd.)

- **Example**: Let's consider a battery which has 10000 mAh of power capacity.

- After dividing 10000 mA hour/1 hour gives 10000 mA = 10 A = **10 C**
- So, a battery with 10000 mAh of capacity will have a C rating of 10 C which means the battery has a capability of delivering 10 A of current at a constant voltage (fixed voltage/rated voltage).

Nominal voltage: While defining power capacity we have a unit called **Wh** which can be elaborated as V * I * hour but where did the voltage gone? As the voltage of the battery will be constant and will not be varied, it is considered as **nominal voltage** (fixed voltage).

So since the voltage is fixed only Ampere and hour are considered as the unit (**Ah/mAh)**.

Charging current: It is the **maximum current that can be applied to charge the battery** i.e., practically maximum of 1A/2A can be applied if a battery protecting circuit is in-built but still 500 mA is the best the range for charging the battery.

Charging voltage: It is the maximum voltage that should be applied to the battery to efficiently charge a battery. Basically 4.2 V is the best/standard charging voltage. Though we apply 5 V to the battery it accepts only 4.2 V.

Discharging current: It is the **current that can be drawn from the battery and is delivered to the load**.

If the current drawn by the load is greater than the rated discharging current, the battery drains very fast which causes the battery heat up quickly which also causes the battery to damage (explode.)

So it is cautious to determine the amount of current drawn by the load as well as the maximum discharging current a battery can withhold.

- **Shelf life:** There might be a situation where the batteries are kept idle/sealed especially in the stores/shops for a long period of time. *So* **shelf life defines the time period a battery can be stay powered up and should be able to use it for a rated time period***.* Shelf life is mainly considered for non-rechargeable batteries because those are of use and throw. For rechargeable batteries even if the shelf time is less, we can still recharge it.
- **Cut-off voltage:** It is the **voltage at which the battery can be considered as fully discharged**, after which if we still try to discharge from it the battery gets damaged. So beyond the cut-off voltage the battery should be disconnected from the circuit and should be charged appropriately.
- **Cycle life:** Let's consider a battery is fully charged and it is discharged to 80% of its actual capacity, then the battery is said to be completed one cycle. Likewise the **number of such cycles that a battery can charge and discharge defines the cycle life**. The more the cycle life the better will be the battery's quality. But if a battery is discharged to say 40% of its actual capacity considering the battery is fully charged initially, it cannot be considered as a cycle life.

Power Density

- It defines power capacity of battery for a **given mass of volume.**
- ***For example:*** 100 Wh/Kg (Alkaline battery standard power density) implies that for 1 Kg of chemical composition it provides 100 Wh of power capacity.
- Now, volume of a **AAA** alkaline battery is 11.5 grams. So if 1Kg can give 100 Wh capacity, then how much a 11.5 gram AAA battery can give??
- Let's calculate Wh (for 11.5 gm) =... **?**
- So, we know the nominal voltage of alkaline battery is **1.5V.**
- Let's calculate Power capacity =... **?**

Energy (Wh)=V× Power Capacity (Ah)

Power Density(Contd.)

Example:

- Let's calculate Wh (for 11.5 gm) = 100*11.5/1000 = 1.15 Wh
- So, we know the nominal voltage of alkaline battery is 1.5V.
- So it provides (1.15/1.5)A * 1 hour gives 0.76 Ah = 760 mAh of power capacity which is almost equal to the power capacity of a standard AAA alkaline battery.

Non-rechargeable Batteries

- These are basically considered as **primary batteries** because they can be used only once. These batteries cannot be recharged and used again. Let's see about the regular, daily life primary batteries that we see.

- **Alkaline batteries:** It is basically constructed with the chemical composition of Zinc (Zn) and Manganese dioxide (MnO_2), as the electrolyte used in it is potassium hydroxide which is purely an alkaline substance the battery is named as alkaline battery having the power density of 100 Wh/Kg.

- **Advantages:**
  - ✔ Cycle life is more
  - ✔ More compatible and efficient for powering up portable devices.
  - ✔ Shelf life is more.
  - ✔ Small in size.
  - ✔ Highly efficient.
  - ✔ Low internal resistance so that discharge state in idle state is less.
  - ✔ Leakage is low.
- **Disadvantages:**

  Cost is a bit high. (& one time use)
- **Applications:**
  - It can be used in torches, remotes, wall clocks, small portable gadgets etc.

Coin cell batteries

- The chemical composition of coin cell batteries is also alkaline in nature.
- Apart from alkaline composition, lithium and silver oxide chemicals will be used to manufacture these batteries which are more efficient in providing steady and stable voltage in such a small sizes.
- It has Power density of 270 Wh/Kg.

- **Advantages:**
  - ✔ Light in weight
  - ✔ Small in size
  - ✔ High density
  - ✔ Low cost
  - ✔ High nominal voltage (up to 3V)
  - ✔ Easy to get high voltages by arranging serially
  - ✔ Long shelf life
- **Disadvantages:**
  - ❖ Needs a holder
  - ❖ Low current draw capability
- **Applications:**
- Used in watches, wall clocks, miniature electronic products etc.

Rechargeable Batteries

- These are generally called as **secondary batteries** which can be recharged and can be reused. Though the cost is high, but they can be recharged and reused and can have a huge life span when properly used and safely charged.

Lead-acid batteries

- It consists of lead-acid which is very cheap and seen mostly in cars and vehicles to power the lighting systems in it.
- These are more preferable in the products where the size/space and weight doesn’t matter. These comes with the nominal voltage starting 2V to24V and most commonly seen as 2V, 6V, 12V and 24V batteries. It has Power density of 7 Wh/Kg.

- **Advantages:**
  - ✔ Cheap in cost
  - ✔ Easily rechargeable
  - ✔ High power output capability
- **Disadvantages:**
  - ❖ Very heavy
  - ❖ Occupies much space
  - ❖ Power density is very low
- **Applications:**
- Used in cars, UPS (uninterrupted Power Supply), robotics, heavy machinery etc..

Ni-Cd batteries

- These batteries are made of Nickel and Cadmium chemical composition.
- Though these are very rarely used, these are very cheap and their discharge rate is very low when compared to NiMH batteries.
- These are available in all standard sizes like AA, AAA, C and rectangular shapes. The nominal voltage is 1.2V, often connected together in a set of 3 which gives 3.6V. It has Power density of 60 Wh/Kg.

- **Advantages:**
  - ✔ Cheap in cost
  - ✔ Easy to recharge
  - ✔ Can be used in all environments
  - ✔ Comes in all standard sizes
- **Disadvantages:**
  - ❖ Lower power density
  - ❖ Contains toxic metal
  - ❖ Needs to be charged very frequently in order to avoid growth of crystals on the battery plate.
- **Applications:**
- Used in RC toys, cordless phones, solar lights and mostly in the applications where price is important.

Ni-MH batteries

- The Nickel – Metal Hydride batteries are much preferable than Ni-Cad batteries because of their lower environmental impact.
- Its nominal voltage is 1.25 V which is greater than Ni-Cad batteries. It has less nominal voltage than alkaline batteries and they are good replacement due to its availability and less environmental impact. The power density of Ni-MH batteries is 100 Wh/Kg.

- **Advantages:**
  - ✔ Available in all standard sizes.
  - ✔ High power density.
  - ✔ Easy to recharge.
  - ✔ A good alternative to alkaline which has almost all similarities and also it is rechargeable.
- **Disadvantages:**
  - ❖ Self-discharge is very high.
  - ❖ Expensive than Ni-Cad batteries.
- **Applications:**
  - Used in all applications similar to the alkaline and Ni-Cad batteries.

Li-ion batteries

- These are made up of Lithium metal and are latest in rechargeable technology.
- As these are compact in size they can be used in most of the portable applications which need high power specifications.
- These are the best rechargeable batteries available. These have a nominal voltage of 3.7V (most commonly we have 3.6V and 7.2V) and have various ranges of power capacity (starting from 100s of mAh to 1000s of mAh). Even the C-rating ranges from 1C to 10C and Power density of Li-ion batteries is 245 TO 430 W/KG.

- **Advantages:**
  - ✔ Power density is very high.
  - ✔ High C-rating.
  - ✔ Cell voltage is high.
  - ✔ Very light in weight.

- **Disadvantages:**
  - ❖ Suffers Aging Effect (even when not in use).
  - ❖ If the terminals are short circuited the battery might explode.
  - ❖ Battery protection circuit is needed.

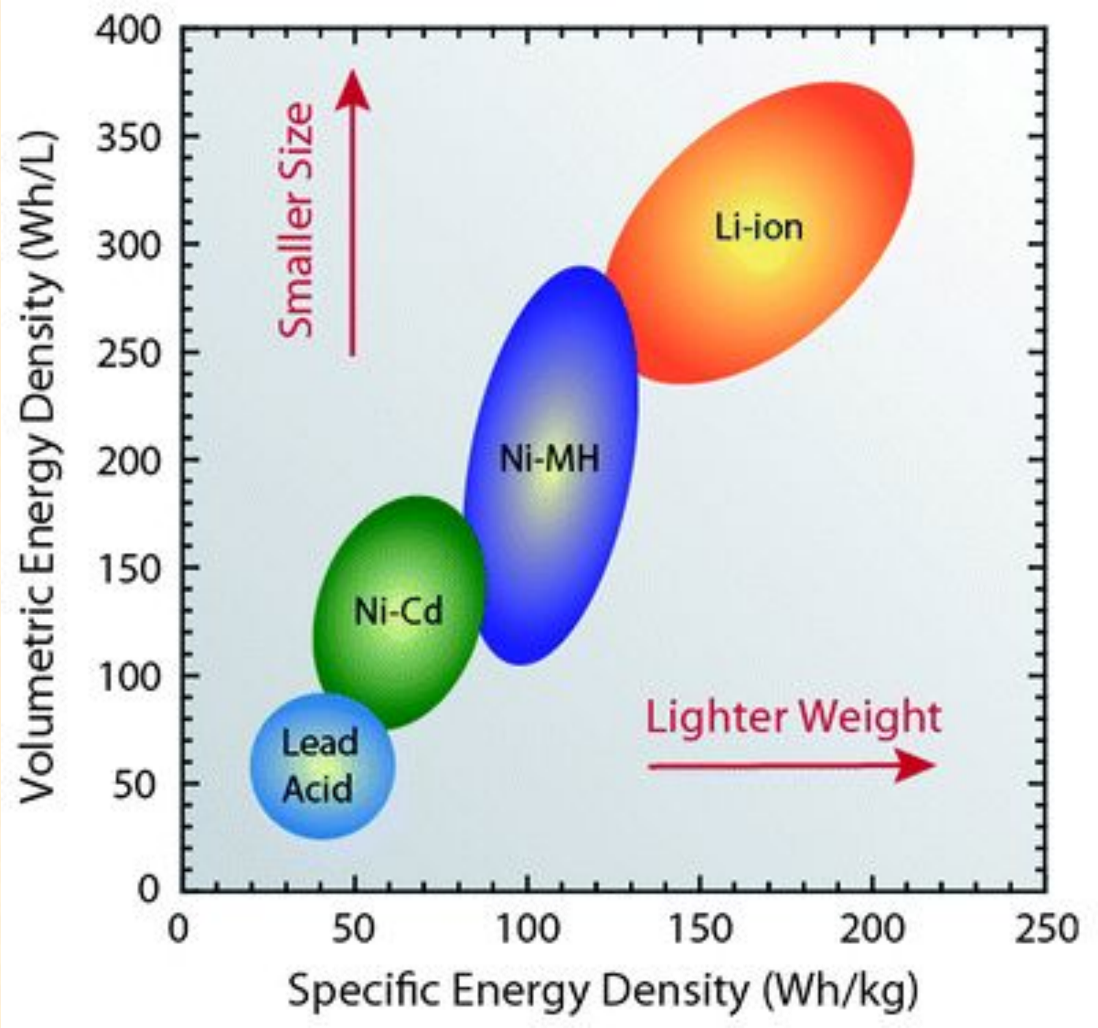
- **Applications:**
  - ✔ Medical tools
  - ✔ Industrial applications
  - ✔ Consumer electronics
  - ✔ Portable devices
  - ✔ Military applications
  - ✔ Power banks

Li-Po batteries

- These are also called as Lithium Ion polymer rechargeable batteries because it uses high conductivity polymer gel/polymers electrolyte instead of liquid electrolyte. These come under the Li-ion technology. These are a bit costly. But the battery is very highly protected when compared to the Li-ion batteries. It has Power density of 245 TO 430 W/KG.

- **Advantages:**
  - ✔ These are highly protective compared to Li-ion batteries.
  - ✔ Very light in weight/ Shape customizable
  - ✔ Thin in structure when compared to Li-ion batteries.
  - ✔ Low discharge rate.
  - ✔ Power density, nominal voltages are comparatively very high compared to Ni-Cad and Ni-MH batteries.
- **Disadvantages**:
  - ❖ Expensive.
- **Applications**: Tablets
  - ✔ Cell Phones
  - ✔ Portable medical devices
  - ✔ Laptops
  - ✔ Compact consumer electronics
  - ✔ Wearable devices
  - ✔ Drones

Energy Density Comparison of Size & Weight



Compare Lithium Ion VS Lithium Polymer

- Advantages
- Disadvantages.

Compare Lithium ION VS Lithium Polymer

Basis	**Lithium Ion**	**Lithium Polymer**
Ageing	Loses actual charging capacity over time	Retains charging capacity better than Lithium =ion
Energy Density	High Energy density	Low as compared to lithium ion
Conversion Rate	The capacity to convert battery into actual power 85-95%	75- 85%
Safety	More Volatile as compared to lithium polymer	More safety. Less chance of explosion
Cost	Cheaper	Slightly Expensive(+30%)
Weight	Heavier	Light Weight
Charging duration	Longer Charge	Comparatively Shorter

Solar-Powered Arduino Precision Agriculture System

Objectives

- To design a **solar-powered embedded system** capable of operating independently of grid power
- To interface and reliably operate a **12V 7-in-1 soil sensor using RS485 communication**
- To implement **energy-efficient operation using duty cycling and low-power modes**
- To ensure **continuous monitoring with at least 2 days of battery backup**
- To optionally enable **IoT-based remote monitoring**

Embedded Systems Power Consumption

- Almost all our day-to-day electrical and electronic devices are designed using embedded systems.

- An embedded system usually comprises of a microcontroller such as ARM, also FPGAs, microprocessors, ASICs and Digital Signal processors along with memory and input and output peripherals.

- Embedded-system based devices are efficient in multitasking, and are also capable of interfacing with the other networks and devices.

- Based on the type of power consumption embedded systems are classified into two types. Portable and Non-portable.

- Portable devices depend on inbuilt battery for power.
- They can be charged and can be used on the go.
- Some examples of portable systems are digital cameras, handheld GPS systems, mobile phones etc. Lithium-ion-pol batteries are the preferred in recent devices due to a highly robust charge discharge cycle.
- Nonportable devices require continuous connection to a source of power. More than the non-portable systems, portable systems are the primary areas of concern, where power management is essential.
- Hence, in a battery-powered system, this need is magnified.

Need for Power Optimization

- **Size and Life of battery.**
  - **Key Aspect**: Extending battery lifespan while maintaining a compact size.
  - **Efficient Power Management**: Ensures devices operate without compromising performance.
  - **Impact on User Experience**: Longer battery life enhances convenience, especially in portable and embedded systems.
- **Performance and Power Utilization.**
  - **Embedded Systems Evolution**: New functionalities are continuously added.
  - **Power Management Importance**: Crucial for maintaining optimal performance.
  - **Balancing Act**: Efficiency depends on managing both power and performance effectively
- **Adaptation of Green Technologies.**
  - **Future Trend**: Shift towards sustainable energy sources like solar power.
  - **Environmental Impact**: Green technologies reduce dependence on traditional power sources.
  - **Carbon Footprint**: ICT is expected to contribute at least 20% of CO_2 emissions, making green energy adoption urgent.

Power Management Techniques

- **Dynamic Voltage and Frequency Scaling (DVFS)**: Adjusts voltage and frequency based on workload to reduce power.
- **Power Gating**: Shuts down unused circuits to minimize leakage power.
- **Clock Gating**: Disables clock signal to inactive modules to reduce dynamic power.
- **Low-Power Modes:**
  - Active Mode: Full performance operation.
  - Idle Mode: CPU halts while peripherals remain active.
  - Sleep Mode: Most system components are turned off except essential parts.
  - Deep Sleep/Hibernate: Minimum power consumption; essential memory retained.
- **Energy Harvesting**: Uses solar, vibration, RF energy for power, ideal for IoT and remote sensors.

Hardware Considerations for Power Management

- **Low-Power Microcontrollers**: ARM Cortex-M series, MSP430 with built-in power management features.
- **Power-Efficient Peripherals**:
    - Low-power communication interfaces (I2C, SPI, BLE).
    - Sensors with ultra-low power requirements.
- **Battery Technologies**:
    - Lithium-ion, Li-Po, supercapacitors for energy storage.
    - Power Management ICs (PMICs) for battery regulation.

Software-Based Power Optimization

- **Operating System Power Management**:
  - RTOS (FreeRTOS) with power-aware scheduling.
  - Dynamic task scheduling to optimize CPU usage.
- **Efficient Coding Practices**:
  - Reduce polling, use event-driven programming.
  - Optimize data transmission to minimize overhead.
- **Power Profiling & Monitoring**:
  - Use tools like Joulescope, EnergyTrace for analysis.
  - Optimize firmware based on power trends.

Power Management in IoT & Wireless Systems

- **Low-Power Communication Protocols**:
  - BLE, Zigbee, LoRa, NB-IoT.
  - Duty cycling techniques reduce power usage.
- **Edge Computing for Energy Efficiency**:
  - Processes data locally to reduce cloud communication.
  - Minimizes transmission power consumption.

How do you Calculate MCU power consumption for low-power applications ?

- ***Power Capacity= V * I * hours = Watt-hour***

How do you Calculate MCU power consumption for low-power applications ?

- To minimize power or energy consumption from the system microcontroller, a developer can simply examine product datasheets to determine the current consumed at the CPU frequency needed for the application.
- ***Power Capacity= V * I * hours = Watt-hour***
- Multiply this current by the battery voltage, and use the resulting data to choose the lowest power microcontroller.
- This seems simple;

However, consider a few hypothetical questions about typical applications to determine if this view of power consumption is comprehensive:

- ***Does the system go into a standby mode when the microcontroller is not running?***
- ***Is the system required to automatically wake itself up at specific time intervals?***
- ***Does the system take any real-world analog signal measurements?***
- ***Does the system need to record any data for analysis or transmission at a later time?***

To gain a comprehensive view of microcontroller power consumption, we must consider these primary power categories:

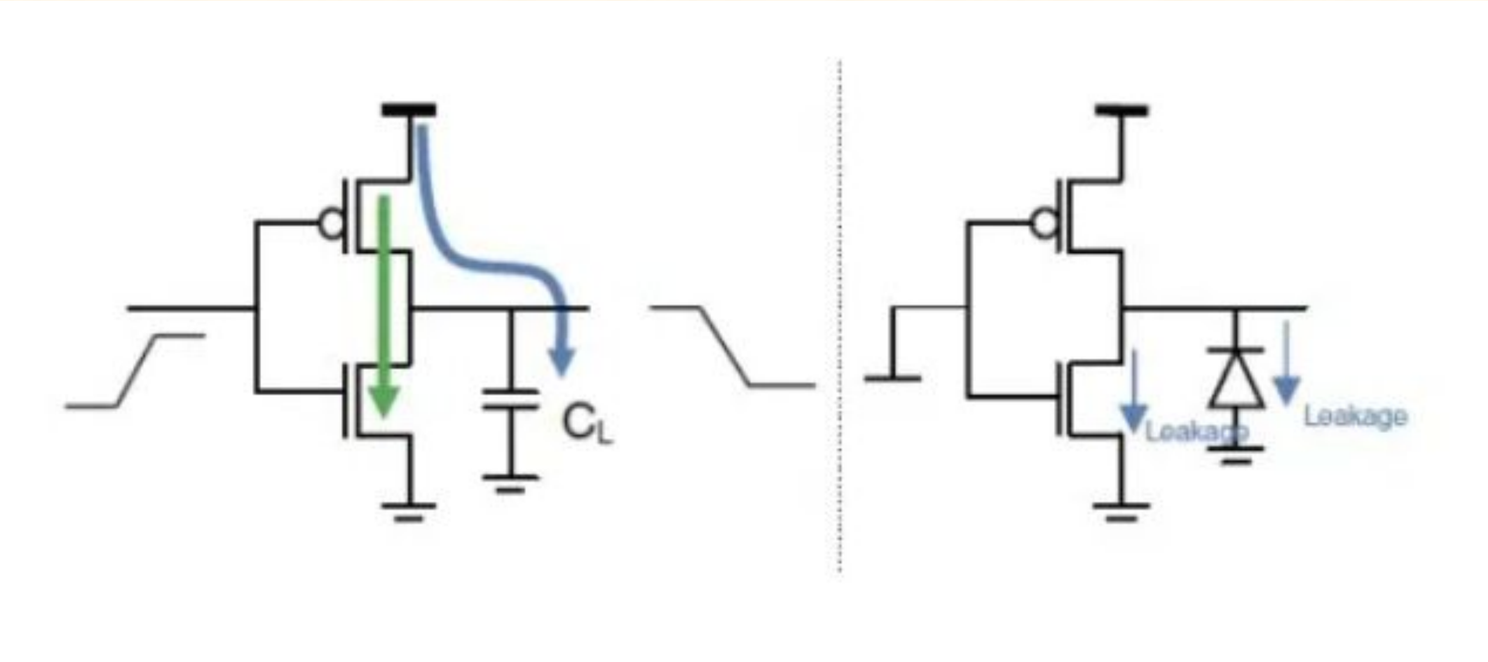
- ***Active power:***

$$P = f\ C_L V_{dd}^2$$

- ***Standby power:***

- ***Transition power:***

CMOS inverter



Total Power = $P_{switching}$ + $P_{short\text{-}circuit}$ + $P_{leakage}$

To gain a comprehensive view of microcontroller power consumption, we must consider these primary power categories:

- ***Active power:***
  - ***Active power consumption by an MCU is the power consumed when the MCU is running. As almost all controllers are based upon CMOS logic, power is consumed primarily during the switching of transistors. The power consumption of a CMOS inverter is the basic building block of any CMOS design.***
- ***Standby power:***
  - ***Majority of product life spent (Waiting for an internal/external event).***
  - ***In many battery based application Standby Power consume major amounts.***
  - ***Standby current is the sum of leakage current, current consumed by power management circuits, clocking systems, power regulators, RTC, IOs, interrupt controllers, and so on.***
- ***Transition power:***
  - ***Finally, the power consumed while transitioning from low power mode to active mode should not be ignored. Devices may end up wasting a significant amount of power while transitioning between these two modes.***

Average Power Consumption

- ***Based on these powermodes, an MCU's average power consumption is:***

$$\textit{MCU power consumption} = \frac{\textit{(Active Power + Sleep Power + Transition Power)}}{\textit{Total Time}}$$

Active Power- *Time for which MCU is active * Active current.*

Sleep Power- *Time for which MCU is I sleep * sleep current.*

Transition Power- *Power consumed while making transition from sleep to active mode*

Power Consumption based on Application

- **Application-dependent: Time spent in mode**

- **Additional modes:** hibernate mode, deep sleep mode **consumes power in the** order of 10s of nA.

- **Majority of time, which power consumption mode:**

- **Vital tradeoffs:**
  - **Beneficial for some system to run at a higher speed.**
  - **Alternatively, slower speed to keep active power consumption low.**

- The active and standby mode time spent is **application-dependent.**
- There are MCUs available on the market which come with additional power down modes other than sleep, such as hibernate mode, deep sleep mode, or shut down mode in which power consumption can be on the order of 10s of nA.
- System designers need to look at the **power consumption of the mode** in which the system has to **operate for the majority of time** to ensure that the overall design is as power efficient as it can be.
- If we look deeper, there are some **vital tradeoffs** that must be considered. For some applications, it could prove **beneficial for the system to run at a higher speed** so it can finish its job faster and return to a low power mode.
- **Alternatively, other systems may do better running at a slower speed to keep active power consumption low.** Here, the system designer has to analyze the best case for the application considering the current at different operating speeds, the time it takes to come out of low power mode, the current consumption in low power mode, and the frequency with which the system needs to switch between active and sleep modes.

Peripheral Power

- MCU power consumption is only one factor when considering system power consumption.

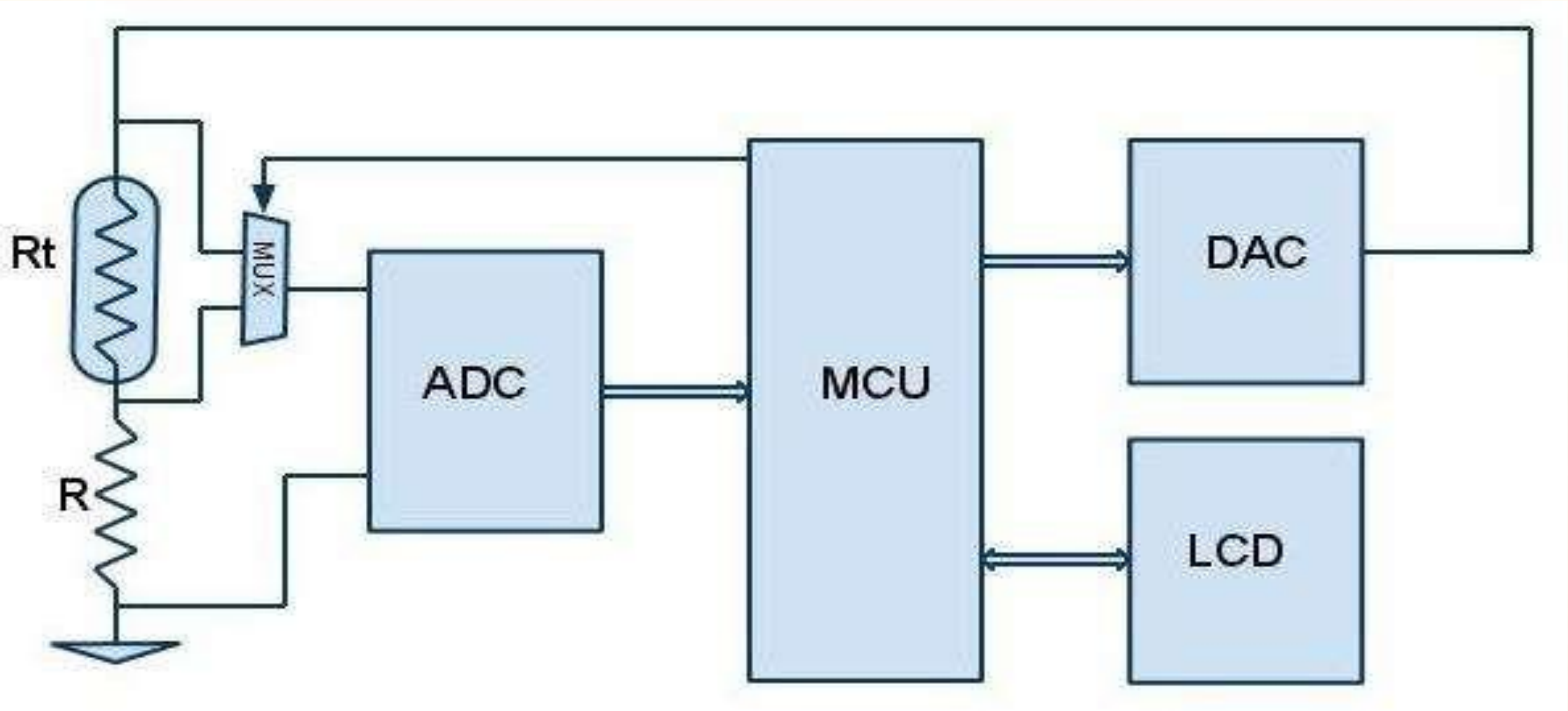
- Sometimes, engineers tend to concentrate too much on the MCU and ignore the power consumed by external peripherals.

- If the objective is to optimize the power consumption of the entire embedded system solution, one cannot afford to do this.

Case: Consider a simple temperature measurement system for home use

- This system has one ADC to measure the sensor voltage.
- DAC to generate a reference signal.
- LCD module to display data, and one MCU to process the data.
- Power should be saved beginning at the individual block level. If the power consumption is calculated for this system, it will be given by:
  - $Total_{Power\ Consumption} = (MCU + ADC + DAC + LCD)_{Power\ Consumption}$

Consider a simple temperature measurement system for home use



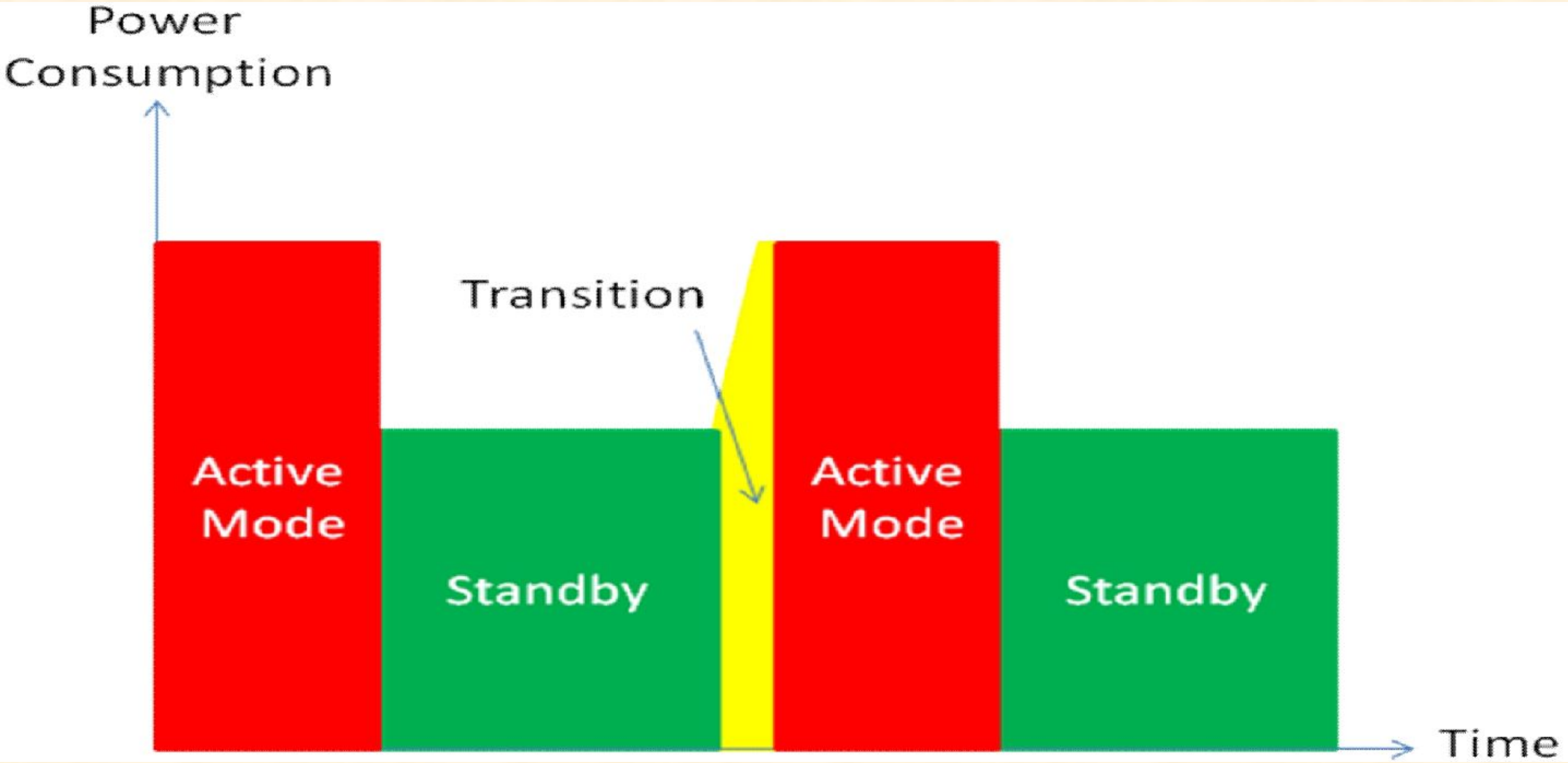
- For this system, **the sample rate need not be very high** since temperature does not change rapidly.

- Power consumption can be kept to a minimum by switching on the ADC and DAC only when required and optimizing the time ratio for which the MCU stays in active and standby mode.

- If this system is made of **discrete components**, it can get quite challenging to coordinate.

- Power consumption for discrete component-based architectures will look similar to the system shown in Figure 3.

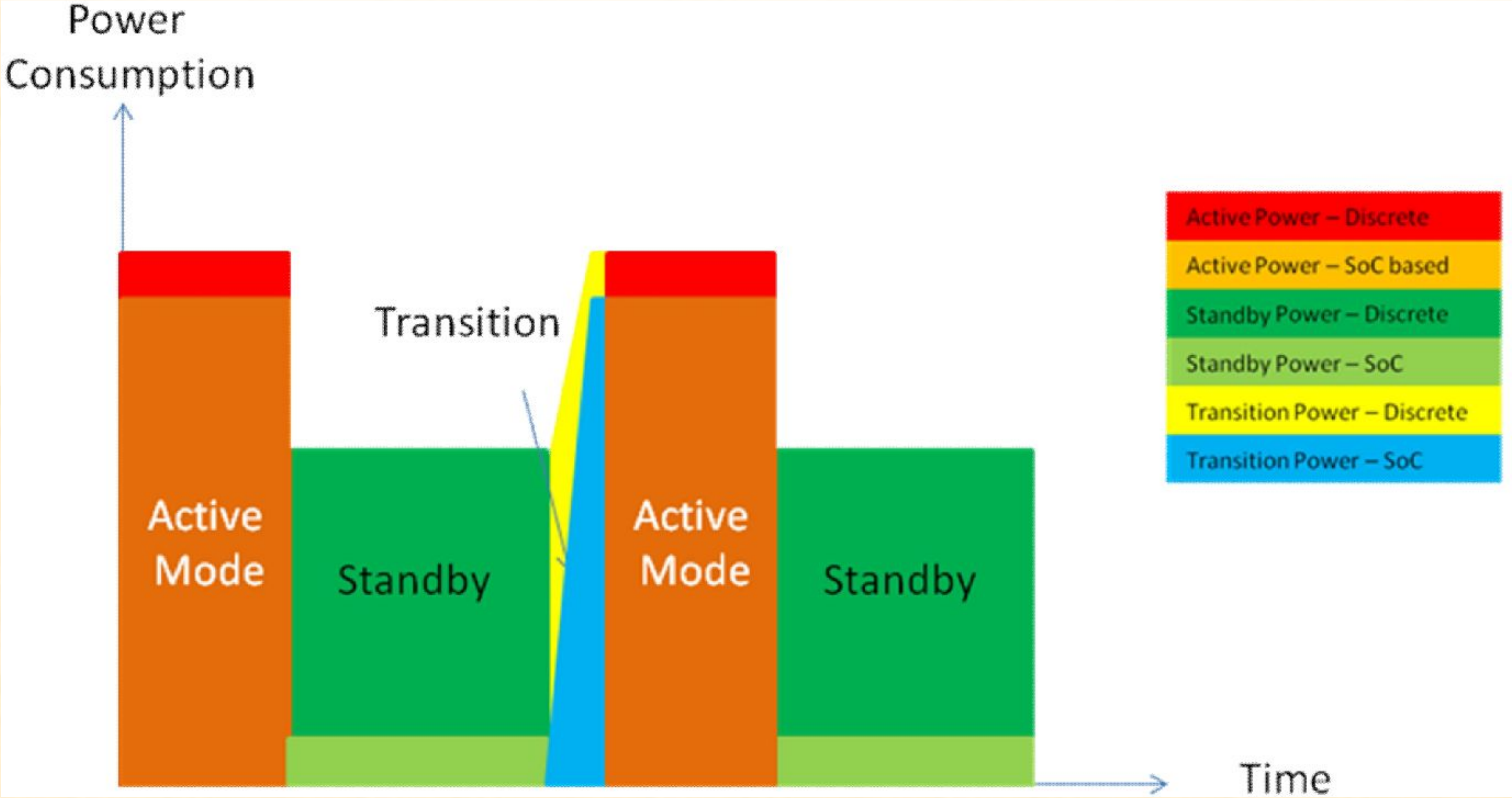
Power consumption system using discrete-based designs



Standby current is mainly contributed to by MCU standby current and the active current of the ADC, DAC and LCD.

The ADC may have an option by which the MCU can stop ADC conversion before it goes to sleep to save ADC power consumption as well. However, there will still be some standby current for ADC and similarly for the DAC.

Power consumption of SoC based solution Vs Discrete solution



- Alternatively, the system could be implemented using a **system-on-chip (SoC)** architecture where all of the peripherals are integrated onto a single chip along with the ability to control the power of each individual peripheral.

- Advancements in SoC technology allow developers to access a wide range of on-chip peripherals such as filters, ADCs, DACs, Op-amps, and programmable analog and digital blocks.

- Modes range from active mode where all the features on the device are available to hibernate modes where current can be as low as 100 nA while retaining the contents of configuration registers and RAM.

- As complex as SoC architectures are, they represent almost the complete system, making it more straightforward to compute power consumption.

- **For example**, if the system is doing nothing, then the standby current of the whole system can be as low as 100nA.

- Since peripherals and the MCU can be switched on or off individually, only the appropriate blocks need to resume operation after the next wake up event.
- This is one of the key features which an SoC can have from a system point of view.

- A system’s average power calculation in a SoC-based system becomes more complex since along with the average MCU current, we need to consider each the operating state of each individual peripheral on the chip. Average system current is:

  - $$\boldsymbol{I_{avg}} = \sum_{i=0}^{no.of\ power\ modes} [(Current\ Consumption\ in\ power\ mode\ * Time)$$
    $$\mathbf{+}\ Current\ Consumption\ in\ Transition\ from\ low\ power\ to\ active\ mode$$
    $$\mathbf{+} \sum_{i=0}^{no.of\ pheripherals-1} (Current\ Consumption\ of\ pheriperal * Time)]/\boldsymbol{Total\ Time}$$